Business-Oriented IT Management: Developing E-Business Applications with E-BPMS

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ABSTRACT
The characteristic feature of e-business applications is the complexity due to the large number of factors that have to be taken into consideration and aligned with regard to the products and services offered, the business processes used, the organization and the information technology (IT) applied. Both researchers and practitioners state, that the alignment between IT and the competitive strategy of the business is paramount. In this paper, we will discuss the E-BPMS modeling framework that takes into account the requirements of a complex e-business application development and we will describe a case from the insurance sector as a proof of concept.

Categories and Subject Descriptors
H.1.0 [**Models and Principles**]: General

General Terms
Algorithms, Management, Measurement, Documentation, Performance, Design, Verification.

Keywords
Complexity Management, E-BPMS, Framework, Business-oriented IT Alignment, Metamodel, Modeling, Service orientation, Key performance indicator

1. INTRODUCTION
Some significant trends like rapid strategic business change, experienced user communities, technology complexity, service oriented architecture [20] and changing requirements regarding the alignment of the IT to the business [22], demand new integrated approaches to building e-business applications. By using frameworks such as the Zachman framework [24], the "IBM Application Framework for e-business" [8] or UML [19] the focus is primarily put on the development of software applications and the technical integration of heterogenous systems. The weakness of these approaches is a missing linkage between the explicit definition of the business objectives and the IT infrastructure, i.e. a framework for a complete and continuous composition of the company architecture in terms of business engineering. This includes the provision of concepts regarding the different layers from business to IT such as strategic aspects, business models, products, user interaction and IT services and infrastructure.

This paper introduces a framework that eliminates the described shortcomings. It proposes a holistic view on the development of service oriented e-business applications by using modeling methods. Section 2 shows the core processes of the framework and the model types based on meta models. Section 3 gives an example of one such meta model with a focus on one sub-process. A brief case study in section 4 illustrates the implementation of the approach in an insurance company. Finally, section 5 offers a conclusion.

2. The E-BPMS Framework
The E-BPMS framework is based on the generic BPMS-Paradigm[1] [16]. E-BPMS extends BPMS in the way that it defines concrete model types for e-business scenarios. Its main difference to existing traditional modeling frameworks is the integrated approach that regards the IT-Management level from a business point of view. It pursues a generic procedure and is not restricted to a certain type of e-business model (e.g. B2B, B2A or B2C). Based on the life cycle oriented BPMS-Paradigm certain processes and corresponding modeling types are distinguished to reduce the complexity and support the construction and demonstration of interdependencies within an e-business application. The five processes are:

1. Strategic Decision Process: In this process two modeling types are used: The first one is the E-Strategy Model. This model type uses the Balanced Scorecard [10,11,12] methodology to support the definition of perspectives, success factors, goals, indicators and initiatives to set targets and to define the strategic positioning of the company within the market. It is the basis for the second model type namely the E-Business model. Here, the typical parts of a business model [23] are considered, such as the dynamic part (e. g. product and service architectures, service level agreements, product and information flows, business actors and their roles), the business benefit part (description of the potential benefits), and the revenues part (sources of revenue).
2. Design Process: In this phase the design of the products and services, the business processes, the organizational

[1] BPMS = Business Process Management Systems

structure and the IT service catalogue, that includes all the requirements for the implementation, takes place. Based on traditional approaches e.g. for business process modeling [14] specific model types are derived for e-business applications (e.g. to take into account the electronic interaction and flow of information between the different actors in a business process).

3. Resource Allocation Process: The definitions derived from the previous phase are transformed into detailed models and then translated into the e-business solution. The models available are the Application Landscape Model for the overview of the applications, the IT Process Model to describe the interaction of the user and the service oriented workflow, the Authorization and Competence Model, the Software Architecture Model, simply showing the IT services as components with their interfaces and data, and the ICT-Infrastructure Model for the planning of the deployment of the selected software architecture.

4. Execution Process: This phase observes the results of the set indicators at run time and stands for the actual execution of the e-business application.

5. Performance Evaluation Process: This process is a very important part to monitor and analyze the gained results and to generate input for the Strategic Decision and Design Process to obtain the closed "double-loop". It reverts to an Evaluations Model that defines how the monitoring of results from the Execution Process takes place and how they are linked to Design and Strategy.

Figure 1 shows the BPMS-sub processes with the corresponding E-BPMS model types.

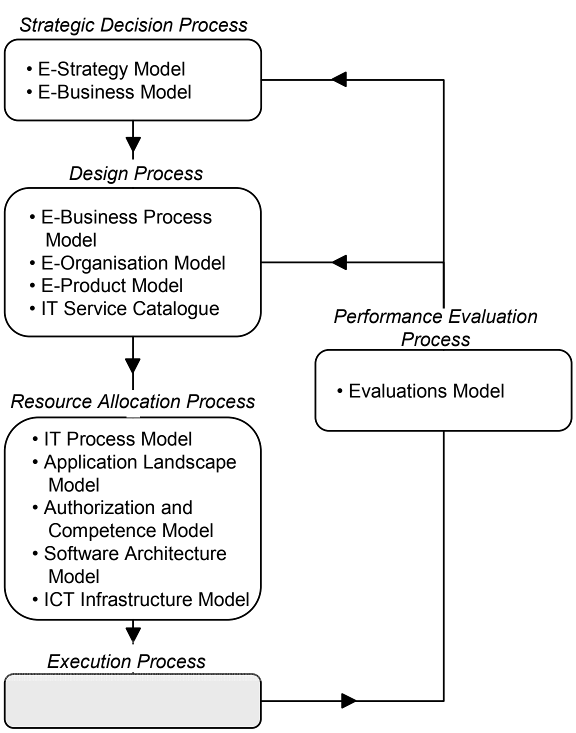

Figure 1. The E-BPMS Modeling Framework [1]

3. The Performance Evaluation Process

For illustrating the details of the model types in the E-BPMS framework the Performance Evaluation Process has been selected. It is an important process to observe the results of the objectives regarding the strategic aspects with the focus on product and services, IT, organization and business processes. The scope is set to certain evaluation items within the different model types to cluster the core elements as input for the monitoring and evaluation.

The model types for E-BPMS are specified in the form of metamodels [13,15]. This allows to define the syntactic linkages between the model elements and provides the foundation for an IT-based implementation. Furthermore, meta models can be used to align the model elements to explicit semantic descriptions, e.g. in the form of ontologies [4,13] and as a starting point for the assignment of visualizations [6,7].

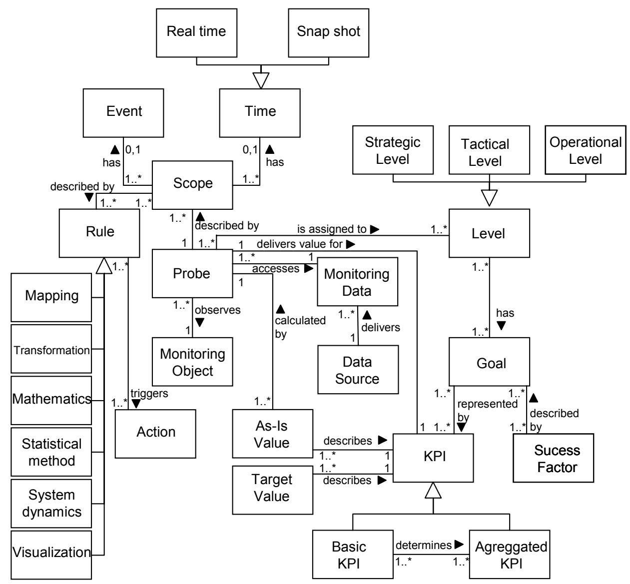

Figure 2. Meta model for service oriented business monitoring [1, 9]

Figure 2 shows a meta model in UML notation for the service oriented business monitoring that has been integrated in the E-BPMS framework. The modeling classes <Success Factor>, <Goal> and <KPI> (Key Performance Indicator) are the linkage to the E-Strategy Model. The modeling class <Probe> is described by the scope and observes the monitoring object (e.g. an activity in an e-business process model). Rules and mechanisms can be defined for the scope like mapping, statistical methods, visualization and so forth, to process and aggregate the data.

Selecting the right key performance indicator set is a crucial aspect. Within the Strategic Decision Process it is often the case that the focus is strongly set on already existing indicators and data. Thereby it is tempting to adapt existing measurement figures to the objectives defined. It is thus important to be flexible in choosing indicators and not to be afraid of defining non existing figures [21]. In the first step the adequate figures that seem best for the measurement of the set objectives should be identified. In the next stage all possible figures are described and analyzed in

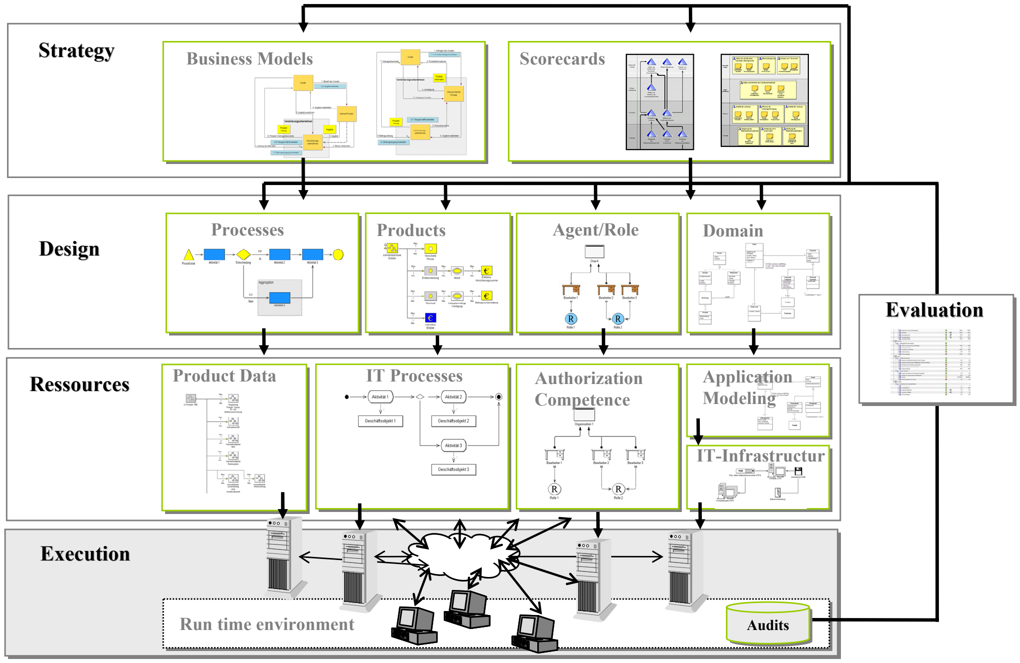

Figure 3. Modeling levels and Model Types

terms of exact meaning, method of calculation, measuring periods and units of measurement. In the last step criteria and requirements are checked to get the best indicator set under economic considerations. Such criteria could be the following [21]:

- Relevance: The indicator should be relevant for the addressee.
- Expressiveness: The meaning for the people using the indicator should be clear and not misleading.
- Cost-Benefit/Feasibility: The ratio between the cost and the benefit of getting and using the indicator should be acceptable.
- Reliability: Repeated measures of the same indicator always deliver the same result.
- Sensitiveness: The indicator should react strongly on changes.

Regarding the indicators used for IT controlling it has been observed that the demand for a strong alignment to business is growing and that IT service management is enhanced with business metrics. A shift from decision making based on technical figures to business metrics should gauge the business impact. Especially regarding the IT services, mostly technical metrics are used such as availability, throughput and response time and decisions are made strongly focused on these IT metrics. But business-oriented management of the IT must help to achieve the set business goals and it should use business indicators to communicate the results [22].

4. B2B INSURANCE APPLICATION

The following description gives a brief insight into a B2B insurance application that used the E-BPMS methodology as the modeling framework and the meta modeling software ADONIS® as the implementation tool. The main idea was to enable insurance companies to offer their services and products via an internet platform, provided by a third party, so that the customers and agents benefit from reduced complexity, faster execution of the processes and the reduction of administrative costs. Besides the general advantages of a modeling approach like simulation algorithms, benchmarks, business activity costs analyses, IT alignment to business goals etc., the following aspects showed a big impact on the result of the application:

- Product and service management: Through the model based design in E-BPMS the big amount of products and services offered by the insurance companies as well as the implications regarding the processes and the software development part could be systematically analyzed and optimized at an early stage.
- Certification and integration of the processes of the different parties: Due to documented processes of the parties and the fulfillment of set criteria such as interfaces, availability, integrity etc. certificates could be obtained.
- Evaluation and test management: Because of the availability of a coherent set of models describing the intended e-business solutions it was possible to simulate the operation of the application prior to its implementation. Thereby different results mapped to changing situations and conditions could be monitored.

Figure 3 shows the different levels based on the BPMS-Paradigm with concrete examples for the E-BPMS model types.

5. CONCLUSION

The paper depicts the E-BPMS framework based on the BPMS-Paradigm for an integrated modeling approach to develop e-business applications. The different views and model types are specified in the form of metamodels and support a continuous business-oriented procedure and the alignment of IT to business. As an example of the underlying metamodels the Evaluation Process was selected and the meta model was introduced using the UML notation. With the practical example for an e-business application for insurances the advantages of the framework could be shown.

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